Today: How do we place poles for the observer?

Observer-based control design - the separation principle

I Exploiting Duality For Pole Placement

Control Design

$P = [\lambda_1 \cdots \lambda_n]$

$K = \text{Place}(A, B, P)$

K is produced s.t.

$$\chi_{A-BK}(\lambda) = \prod_{i=n}^{n} (\lambda - \lambda_i)$$

Products of

Observer Design

$$\chi_{A-LC}(\lambda) = \prod_{i=1}^{n} (s - s_i)$$

A) Duality Between Controllability and observability Matrix

$\Gamma = [B \;\; AB \;\; \cdots \;\; A^{n-1}B]$ $\quad \Omega = [C \;\; AC \;\; \cdots \;\; A^{n-1}C]^T$

This relation is also in the characteristic Equation

$\det(M) = \det(M^T)$ if M is square

$$\Rightarrow \chi_A(\lambda) = \det(\lambda I - A) = \det(\lambda I - A^T)$$

$$\Rightarrow \chi_{A-LC}(\lambda) = \det(\lambda I - (A - LC)) = \det(\lambda I - (A - LC)^T)$$

$$= \det(\lambda I - (\underbrace{A^T}_{\hat{A}} - \underbrace{C^T}_{\hat{B}} \underbrace{L^T}_{\hat{K}}))$$

Recall $\chi_{A-BK}(\lambda) = \det(\lambda I - A - BK)$

Transpose fixes the ordering

So now if A, C is C.O. then (A^T, C^T) is C.C. For

$$\dot{z} = A^T z + C^T u$$

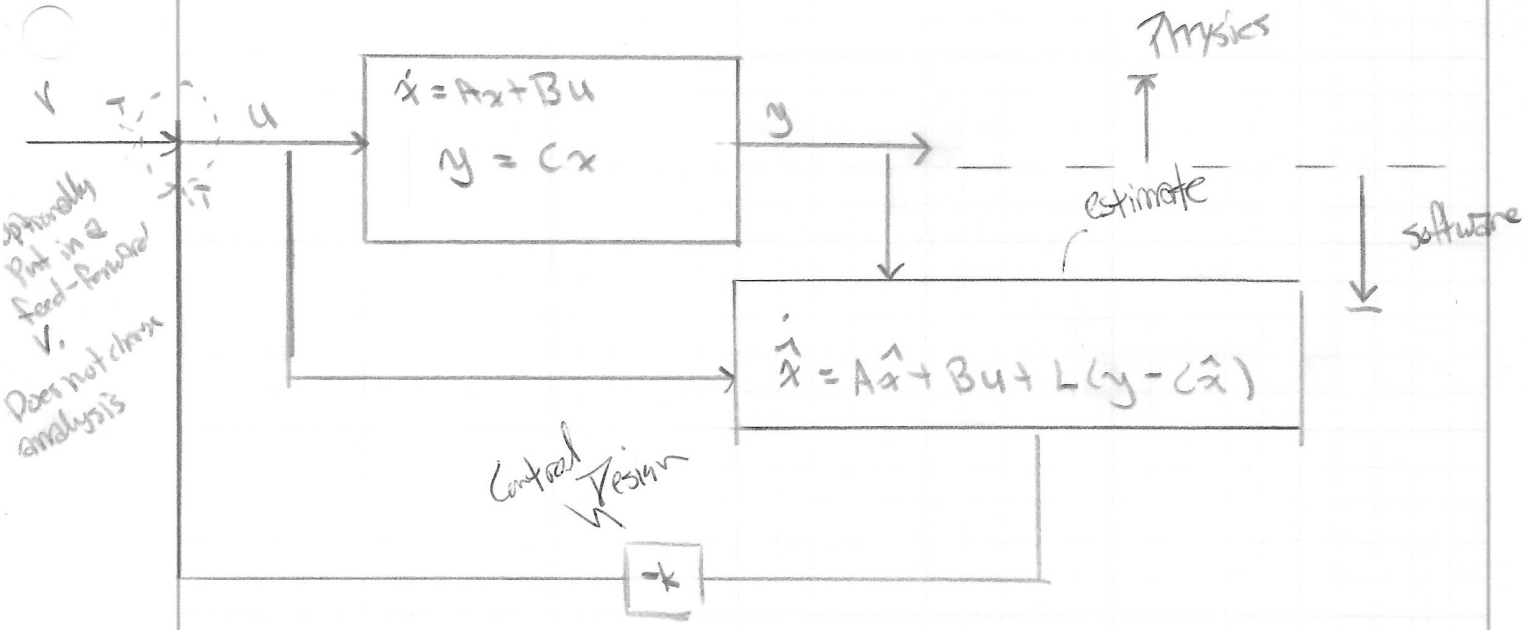
Therefore we can use the same commands in MATLAB to do observer design

$$P = [s_1 \cdots s_n]$$

$$L = (\text{place}(A^T, C^T, P))^T$$

Observer Based Control Design

Key Idea: $u = -k\hat{x}$ ⟷ Full state feedback using estimated state



A) State Dynamics

$$\dot{x} = Ax - Bk\hat{x} \quad ; \quad \dot{\hat{x}} = A\hat{x} - Bk\hat{x} + LC(x - \hat{x})$$

This is is easier to evaluate if we look at

$$\dot{x} \;\&\; e = x - \hat{x} \Rightarrow \dot{e} = \dot{x} - \dot{\hat{x}} = Ax - \cancel{Bk\hat{x}} - A\hat{x} + \cancel{Bk\hat{x}} - LC(x - \hat{x})$$

$\Rightarrow \underline{\dot{e} = (A - LC)e}$

$$\dot{x} = Ax - BK\hat{x}$$
$$\Rightarrow \dot{x} = Ax - BK(x - e)$$
$$\Rightarrow \dot{x} = Ax - BKx + BKe$$
$$\Rightarrow \underline{\dot{x} = (A - BK)x + BKe}$$

III Aggregate Dynamical System

$$z = \begin{bmatrix} x \\ e \end{bmatrix} \Rightarrow \dot{z} = \begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix}$$

$$\Rightarrow \dot{z} = \begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - BK & -BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix}$$

Recall for a block triangular matrix, the eigenvalues are those of the block diagonal elements

↳ The system matrix will be a stability matrix as long as

True because of this

1) $A - BK$ is a stability matrix
2) $A - LC$ is a stability matrix

} Trivial if C.C & C.O.

A) Separation Principle: We can design the control and observer seperately

IV Separation Principle

Typically in practice, the observer is desired to be faster than the controller by placing poles of $(A - LC)$ far to the left of poles from $(A - BK)$

↳ Good rule of thumb: We want an order of magnitude of difference.

Caveat: Larger the magnitude of observer poles requires better models.

I Detectability

Controllability

Three possible results

- Controllable

  $\exists K$ s.t. the eigenvalues $A - BK$ can be arbitrarily located

- Stabilizeable

  $\exists K$ s.t. $Re(\lambda) < 0$ $\forall \lambda \in eig(A - BK)$

- Not stabilizeable

  $\nexists K$ s.t. $Re(\lambda) < 0$ $\forall \lambda \in eig(A - BK$

Eigenvalue Tests

- Controllable

  iff $rank([\lambda I - A \;\; B]) = n$ for each $\lambda \in eig(A)$

  iff $rank(\Gamma) = n$

- Stabilizeable

  iff $rank([\lambda I - A \;\; B]) = n$ for each $\lambda \in eig(A)$ s.t. $Re(\lambda) > 0$

  or evaluate eigenvalues of uncontrollable portion

Observability

Three possible results

- Observable

  $\exists L$ s.t. $eig(A - LC)$ can be located arbitrarily

- Detectable

  $\exists L$ s.t. $Re(\lambda) < 0$ $\forall \lambda \in eig(A - LC)$

- Not Detectable

  $\nexists L$ s.t. $Re(\lambda) < 0$ $\forall \lambda \in eig(A - LC)$

Controllable

Eigenvalue Tests

iff $rank\left(\begin{bmatrix} \lambda I - A \\ C \end{bmatrix}\right) = n$ for each $\lambda \in eig(A)$

iff $rank(\Omega) = n$

- Detectable

  iff $rank\left(\begin{bmatrix} \lambda I - A \\ C \end{bmatrix}\right) = n$ for each $\lambda \in eig(A)$ s.t. $Re(\lambda) > 0$

  or eigenvalues of unobservable portions